

HIGH-RESIDUE CONSERVATION TILLAGE FOR ROW CROPS

Although the conservation tillage unit at the NCSU Thompson Farm is currently on hiatus, a dedicated group continues to dig into this critical area of research. We are busily engaged in tactical operations behind the wire, and this is the first installment of what we hope will become an intermittent series of dispatches from the “underground” – reports on ameliorating the crop rooting environment.

The conservation tillage research initiative at the Center for Environmental Farming Systems (CEFS) seeks innovative ways to remove soil physical and chemical barriers to root growth impacting agro-productivity. In other words, how do we create a soil environment enabling crops to express their full yield potential? This is, in a nutshell, the fundamental concern of soil management. To ameliorate crop stress we must employ an integrated program addressing soil erosion, crusting, compaction, moisture retention and storage, carbon fixation, and fertility. Tillage, in our view, is a disruptive and energy-consuming operation that should be deployed precisely to relieve targeted zones of soil compaction and for seedbed preparation where needed.

Our research group also looks for ways to reduce off-farm inputs as far as may be practical without sacrificing productivity. On the other hand, we accept that modern agriculture relies on a diverse array of technologies and approaches to be successful and that, ultimately, it's the farmer who must decide upon the best mix. Our research may be described as adaptive, where no *modus operandi* is taboo provided it boosts productivity, quality, income, sustainability, and is consistent with the principles of good land husbandry. In short, we are mavericks looking for solutions wherever, and in whatever guise they may be found.

Our current focus is on evaluating the mechanical roller-crimper as a residue management tool concurrently at CEFS and at the Upper Piedmont Research Station (Figure 1).

Mechanical rollers have long been employed by farmers in Brazil, Argentina, and Paraguay to successfully manage high density cover crop residues in production systems using the guiding principles of ‘zero’ and ‘minimum’ tillage (Derpsch, 1998; Derpsch et al., 1991).

Basically, the roller-crimper relies on the ballast of a cylindrical drum to flatten and crimp standing cover crops, leaving a pressed, intact blanket of soil protective mulch oriented in the direction of planting. This has been termed high-residue conservation tillage (Lee et al, 2002; Reiter et al, 2002; Torbert et al, 2002).

Although a precise definition of ‘high-residue conservation tillage’ has not been coined, it is defined herein as an agricultural production system consisting of: (1) limited or no-tillage; and (2) further exploiting crop and cover crop residue to maintain full soil cover before canopy closure. Initially, interest in the high residue concept was spurred by observations from soil nitrogen recovery studies with small

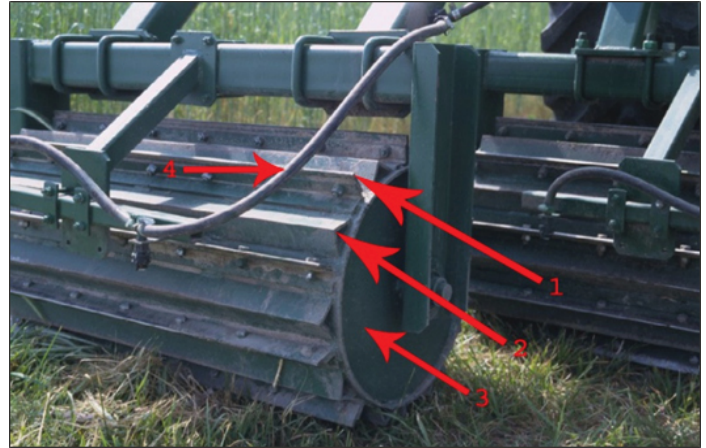


Figure 1. Mechanical roller-crimper built by Kelly Manufacturing Company (KMC), Tifton, GA. Arrows show principal parts of the roller: (1) crimper blade; (2) damping iron to reduce vibration; (3) drum roller w/end plug (not shown) allows user to fill drum with water to increase down pressure on crimper blade; (4) plumbing assembly to apply burndown herbicide.

grains at the Thompson Farm over the period 1999-2002. As happened, some of the small grain plots were left undisturbed, allowing the plants to mature and subsequently fall over and die. Wherever this had occurred, a perched canopy of desiccated small grain covered the soil surface. Peering beneath this canopy one could see few if any weeds germinated as late as the first week of July. In contrast, plots where the grain plants had been cut and removed were flush with weeds. The question naturally arose: could this effect be duplicated in row crops?

About the same time, visits to Latin America by researchers from the National Soil Dynamics Laboratory in Auburn, Alabama, brought home news of hand-crafted residue rollers popular with some farmers in Brazil. Prototype rollers were built and field tested in Alabama. Results were encouraging, but (at the time) limited in scope. The Alabama studies begged other questions: Could roller-crimpers be deployed to create a weed-suppressive shield of cover crop residue at the soil surface? Could this residue suppress early-season weeds and provide soil-protective benefits at the same time?

The idea of high-residue conservation tillage is not new. Gardeners who use the deep-mulch method have been practicing this long before the invention of cover crop rollers. Our goal is to extend the deep mulch concept to address the problems of soil carbon loss, soil crusting and compaction, and reduction in the risk of short-term droughts in row cropping systems in the southeastern USA. There's still much that we don't know about the overall impact of high-residue conservation tillage systems on long-term soil moisture balance, fertility, weed, disease, and insect management. Following is a brief overview of the objectives, methods, and early results obtained from NCSU's first mechanical roller in 2004. A full 2-year summary of this work is available [here](#).

Objectives

The immediate goals of this project were to evaluate weed suppression, residue management, and soybean and cotton lint yield response in a high-residue conservation tillage system to test: (1) the physical effect of surface pressed, intact residue and residue orientation on early-season weed suppression using different weed control programs; (2) the relationship between residue decomposition and incident weed pressure; and (3) the effect of residue management on soybean and cotton stand establishment, growth, and yield.

Methods

Research was initiated in fall 2003 by establishing a small grain winter annual cover crop (rye cv. 'Abruzzi') at each of two sites with similar weed management histories, field C-9 (cotton) and field C-10 (soybean). Soil at both sites was nearly level, well-drained and moderately permeable Wickham sandy loam. Prior to rye growth termination, residue cover was estimated using 0.5 m² quadrats. Rye growth was terminated mechanically or with glyphosate at 1.75 L/ha, and the standing rye residue flattened using one pass of the roller-crimper (Figure 2). The mechanical roller was built to our specifications by the Kelly Manufacturing Company, Tifton, GA. in 2003.

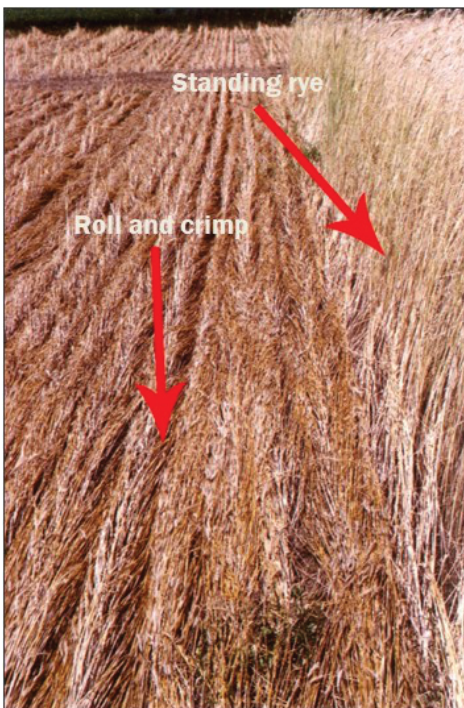


Figure 2. Winter annual rye cover crop after one pass with the KMC mechanical roller. Residue pressed on the surface and oriented in the direction of planting may aid growers in managing high density residue loads.

The roller drum was 46 cm in diameter x 203 cm wide. Crimper blades, consisting of mild steel angled 90° with dimensions of 0.635 cm in thickness x 5.7 cm height, were mounted 18 cm apart around the circumference of the drum. The drum was attached to the toolbar via parallel linkages, which in turn attached to the tractor via a category II 3-point hitch. Optional tank and spray brackets were included but not used in this application. Before use, the drum was filled approximately 1/2 full of water via two threaded plugs installed on either end of the drum.

Full-season soybean (Pioneer 95B97) was planted on 14 May 2004 using a six-row John Deere Maximerge vacuum planter calibrated at 334,300 seeds ha⁻¹ (8 seeds/ft) on 76-cm (30 inch) rows. Cotton (DP 451) was planted on 17 May 2004 using a four-row planter calibrated at 119,170 seeds ha⁻¹ (3.5 seeds/ft) on 96-cm (38-inch) rows. Weed management programs included: (1) rye residue + no herbicide; (2) rye + glyphosate only for burndown; (3) rye + glyphosate + pre-emergent herbicide; (4) rye + glyphosate + pre + post emergent herbicide. All plots were planted no-till except in cotton, where a rip-strip treatment was included. A clean-till treatment and one no-till treatment that excluded the roller, was used for comparison in both trials.

Weed counts were taken at planting and at 2 and 4 weeks post-emergence at three points along a diagonal transect in each plot. Total weed biomass was then estimated at lay-by. Residue decomposition was evaluated by placing folded, intact residue in 2-mm nylon mesh bags at rates equivalent to field conditions and retrieved at 2, 4, 6, 8, and 16 weeks after planting. Plots were also sampled for soil moisture to 45 cm deep in the untrafficked interrow. A randomized complete block design with four replications was used for statistical analysis of data in both trials.

Results

In 2004 cotton (Figures 3 and 4) and soybean (Figure 5) stands were successfully established in the rolled cover crop residue. Stands for both crops were reduced in the rolled residue mainly due to lifting of the planter's gauge wheels in places where the residue was unevenly distributed. Rye residue production averaged 4.97 Mg/ha (4,420 lbs/acre) in cotton and 7.01 Mg/ha (6,253 lbs/acre) in soybeans. Mechanical rolling alone was not as effective at terminating rye growth as glyphosate. When good weed management was achieved, soybean and cotton lint yields in the rolled treatments were about equal to conventional clean-till. However, reducing herbicide inputs resulted in a yield penalty. Weed surveys conducted at two and four weeks after planting showed early establishment of several important weed species where herbicide inputs were either reduced or eliminated despite a lack of soil disturbance and heavy mulch cover.

Decomposition of the rye residue followed an exponential decay function with approximately 70% of the original residue decomposed at 16 weeks after planting.

Overall, highest lint yield was obtained by rolling and ripping combined with pre-and post-emergence weed control. Despite above-average rainfall during the growing season in 2004, sub-soiling a Wickham sandy loam combined with weed control increased cotton lint yield by nearly $\frac{1}{2}$ bale per acre over no-tillage alone.

Interpretive Summary

Early results indicate that row crops can be adapted successfully to high residue conservation tillage systems using the mechanical cover crop roller. We observed that neither high residue density (2-3 tons/acre) nor residue orientation suppressed annual or seedling perennial weeds enough to counteract the need for supplementary early-season weed management using herbicides, cultivation, or a combination. Overall, weed germination was encouraged wherever the ground was disturbed, including such minor breaches caused by wheel traffic and row markers. As shown in Figure 6, it's difficult to completely avoid soil and residue disturbance even under so-called 'zero tillage' conditions.



Figure 3. No-till cotton directly planted in mechanically rolled rye residue.



Figure 4. Strip-tilled cotton, early post-emergence in mechanically rolled rye residue.



Figure 5. Soybean directly planted in mechanically rolled rye.



Figure 6. Weed germination in the furrow cut by Maximerge row marker.

A Final Note

High-residue conservation tillage represents a small but growing worldwide trend toward cropping intensification to cultivate yield gains while reducing in-season weed pressure by extending the period of soil mulch cover. Our group is working in tandem with farmers, researchers, and extension personnel throughout the southeastern USA who recognize the benefit of squeezing pests off the production stage using innovative management approaches. When it comes to tinkering, farmers often lead the way (NRCS, 2002). Scientists and technical staff at places such as the National Soil Dynamics Lab in Auburn, Alabama, Rodale Institute, and the Natural Resource Conservation Service (NRCS) also are forging ahead with high-residue conservation tillage research.

Alternative roller design is one critical area of work that has come under scrutiny by Kornecki *et. al.* (2005) at Auburn's Soil Dynamics Lab. As the photo gallery on page 5 points up, there's a lot of inherent flexibility in the design of mechanical rollers. However, keep in mind the tool has two principal design goals: (1) effective cover crop growth termination; and (2) efficient management of high density residue without imposing additional demands on the grower. Experience is likely to point to no one particular design that fits all situations. Rather, growers are likely to adapt the mechanical roller to their specific cover crop, cropping system, and field equipment situations.

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Further Reading

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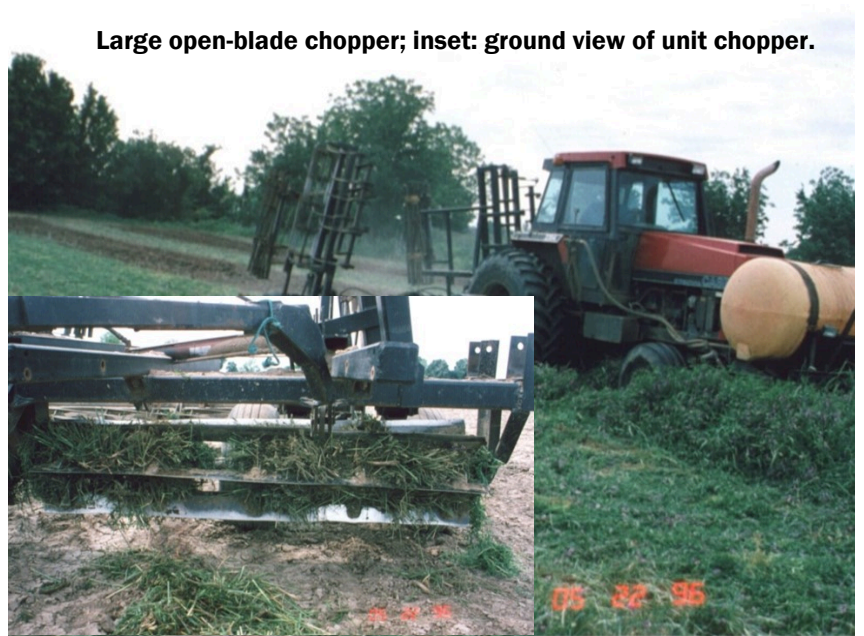
Early drum-type roller built by NSDL, Auburn, AL.



Open blade Buffalo stalk chopper modified by Steve Groff, Cedar Meadow Farm, Lancaster County, PA.



Push roller for small farm or garden.



Large open-blade chopper; inset: ground view of unit chopper.

Photo + inset courtesy of Seth Dabney, USDA-ARS Sedimentation Lab, Oxford, MS.



South American drum roller.



South American drum roller detail.